Netflix Customer Churn Prediction Using Machine Learning

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Abstract— The growth of online streaming platforms has made customer retention an important challenge for subscription-based businesses. Predicting whether a customer is likely to discontinue a subscription can help businesses understand customer behaviour and develop effective retention strategies. This project presents a machine learning approach for predicting customer churn using a Netflix-themed customer dataset. The proposed workflow includes data cleaning, duplicate removal, exploratory data analysis (EDA), categorical data encoding, feature scaling, and train-test splitting. Five machine learning classification algorithms, namely Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, and Support Vector Machine (SVM), are implemented and compared. Model performance is evaluated using accuracy, precision, recall, F1-score, and confusion matrices. In addition, K-Means clustering is applied to group customers into clusters and examine their churn patterns. The project demonstrates how machine learning techniques can be used to analyse customer data, identify churn patterns, and support data-driven customer retention decisions. The performance of each model is compared using the experimental results obtained from the implementation.

Keywords— Customer Churn, Netflix, Machine Learning, Classification, Exploratory Data Analysis (EDA), Logistic Regression, Random Forest, Model Evaluation, K-Means Clustering.

I. INTRODUCTION

The rapid growth of online streaming platforms has changed the way people access and enjoy digital entertainment. Platforms such as Netflix provide subscription-based services and generate large amounts of customer data related to viewing habits, subscription types, login activity, payment methods, and content preferences. Analysing this data helps businesses understand customer behaviour and improve the quality of their services. One of the major challenges faced by subscription-based platforms is customer churn, which occurs when a customer discontinues their subscription.

Machine learning provides effective techniques for analysing customer data and identifying patterns associated with churn. By using customer-related attributes such as age, gender, subscription type, watch hours, last login days, region, device, monthly fee, payment method, and average daily watch time, machine learning models can learn patterns that may help predict whether a customer is likely to leave the service. These predictions can support businesses in identifying customers who may need additional attention and developing strategies to improve customer retention.

This project, titled **"Netflix Customer Churn Prediction Using Machine Learning,"** uses a Netflix-themed customer dataset to develop and evaluate machine learning models for churn prediction. The project involves data preprocessing, duplicate removal, handling missing values, exploratory data analysis (EDA), categorical data encoding, feature scaling, model training, and performance evaluation. Five classification algorithms are considered: Logistic Regression, K-Nearest Neighbours (KNN), Decision Tree, Random Forest, and Support Vector Machine (SVM).

The performance of these models is compared using accuracy, precision, recall, F1-score, and confusion matrices. In addition, K-Means clustering is applied to group customers based on their characteristics and examine the relationship between customer clusters and churn patterns. The main aim is to identify a suitable classification model for customer churn prediction and understand how machine learning techniques can support customer analysis and retention planning. The results demonstrate the practical application of machine learning in analysing customer behaviour and supporting data-driven decisions.

II. OBJECTIVES

The main objective of this project is to analyse customer data and apply machine learning techniques to predict customer churn on a Netflix-themed streaming platform. The project aims to identify patterns in customer behaviour, compare different classification algorithms, and understand how machine learning can support customer retention strategies. The specific objectives are as follows:

1. **To analyse the dataset:** Examine customer information to understand subscription types, viewing habits, login activity, payment methods, and customer characteristics. To perform data preprocessing: Identify and handle missing values, check duplicate records, examine invalid values, and detect potential outliers to improve data quality.

2. **To perform exploratory data analysis:** Identify and remove duplicate records, handle missing values, and prepare the dataset for machine learning.

3. **To perform exploratory data analysis:** Investigate customer characteristics and churn distribution using statistical analysis and visualisation techniques.

4. **To prepare data for model training:** Encode categorical variables, scale numerical features, and divide the dataset into training and testing sets.

5. **To implement machine learning algorithms**: Apply Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, and Support Vector Machine (SVM) to predict customer churn.

6. **To evaluate model performance:** Compare the five classification algorithms using accuracy, precision, recall, F1-score, and confusion matrices.

7. **To identify the best-performing model:** Analyse the experimental results to determine which classification algorithm achieves the highest performance on the dataset.

III. DATASET DESCRIPTION

The dataset used in this study is a Netflix-themed customer dataset containing information about customer demographics, subscription details, viewing habits, and account activity. It is used to analyse customer behaviour and identify patterns associated with subscription cancellation. The dataset supports exploratory data analysis and the development of machine learning models for customer churn prediction.

The dataset contains **14 attributes**, including customer identification, age, gender, subscription type, watch hours, last login days, region, device, monthly fee, churn status, payment method, number of profiles, average watch time per day, and favourite genre. These attributes provide useful information for understanding customer characteristics and their relationship with churn behaviour.

A. Dataset Attributes

The dataset contains the following attributes related to customer information and streaming activity:

- **Customer ID:** A unique identifier assigned to each customer.
- **Age:** The age of the customer.
- **Gender**: The gender category recorded for the customer.
- **Subscription Type:** The subscription plan selected by the customer.
- **Watch Hours:** The total number of hours the customer has watched content**.**
- **Last Login Days**: The number of days since the customer's last login.
- **Region:** The geographical region associated with the customer.
- **Device:** The device used by the customer to access the streaming platform.
- **Monthly Fee:** The monthly subscription fee paid by the customer.
- **Churned:** The target variable indicating whether the customer has discontinued the subscription.
- **Payment Method:** The payment method associated with the customer's subscription**.**
- **Number of Profiles:** The number of profiles associated with the customer account.
- **Average Watch Time per Day:** The average daily viewing time of the customer.
- **Favourite Genre:** The customer's preferred content genre.

These attributes are used to explore customer behaviour and prepare the data for machine learning. The **churned** attribute serves as the target variable for classification, while the remaining relevant attributes are used as input features. Customer ID is excluded from model training because it is an identifier rather than a meaningful behavioural feature.

Dataset Link:

https://www.kaggle.com/datasets/zeyadmohamed26/netflix-customer-churn-and-engagement-analytics

IV. PREPROCESSING AND EXPLORATORY ANALYSIS

The preprocessing stage involved checking the dataset structure, data types, missing values, and duplicate records. Missing numerical values were replaced with the mean, while categorical values were filled using the mode. Duplicate records were removed, and exploratory data analysis was performed using statistical summaries and visualisations to understand customer characteristics and churn patterns.

The dataset was further analysed to identify missing values and unusual observations. Numerical missing values were handled using the mean, while categorical missing values were replaced with the mode. Duplicate records were removed to improve data quality.

Exploratory data analysis was performed using visualisations such as a churn distribution plot, age distribution histogram, and correlation heatmap. These visualisations helped examine customer characteristics and understand patterns associated with customer churn.

Fig.1 Output showing missing values and duplicate records before and after data preprocessing.

```
Missing Values Before Cleaning:
customer_id              0
age                      0
gender                   0
subscription_type        0
watch_hours              0
last_login_days          0
region                   0
device                   0
monthly_fee              0
churned                  0
payment_method           0
number_of_profiles       0
avg_watch_time_per_day   0
favorite_genre           0
dtype: int64

Duplicate Records Before Cleaning:
0

Missing Values After Cleaning:
customer_id              0
age                      0
gender                   0
subscription_type        0
watch_hours              0
last_login_days          0
region                   0
device                   0
monthly_fee              0
churned                  0
payment_method           0
number_of_profiles       0
avg_watch_time_per_day   0
favorite_genre           0
```

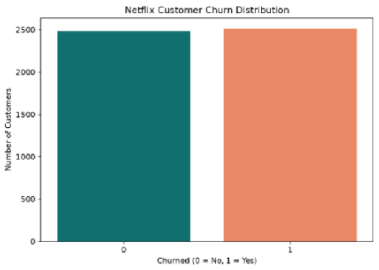
V. DATA VISUALIZATION

Data visualization was performed to understand customer characteristics and identify patterns related to customer churn. A churn distribution graph was used to compare customers who discontinued their subscriptions with those who remained. A histogram was used to examine the distribution of customer ages, while a correlation heatmap was used to study relationships between numerical attributes. These visualizations help in understanding the dataset and identifying patterns useful for customer churn prediction.

A. Churn Distribution

```python
# GRAPH 1: CHURN DISTRIBUTION
plt.figure(figsize=(7, 5))
sns.countplot(data=df, x="churned", hue="churned", palette=["teal", "coral"], legend=False)
plt.title("Netflix Customer Churn Distribution")
plt.xlabel("Churned (0 = No, 1 = Yes)")
plt.ylabel("Number of Customers")
plt.tight_layout()
plt.show()
```

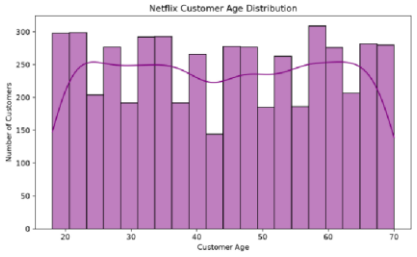
Output:



B. Customer Age Distribution: The customer age distribution histogram is used to analyse the age groups of customers in the dataset. It helps visualise how customer ages are distributed and identify the age groups represented in the streaming platform

```python
# GRAPH 3: CUSTOMER AGE DISTRIBUTION
plt.figure(figsize=(8, 5))
sns.histplot(data=df, x="age", bins=20, kde=True, color="purple")
plt.title("Netflix Customer Age Distribution")
plt.xlabel("Customer Age")
plt.ylabel("Number of Customers")
plt.tight_layout()
plt.show()
```

Output:

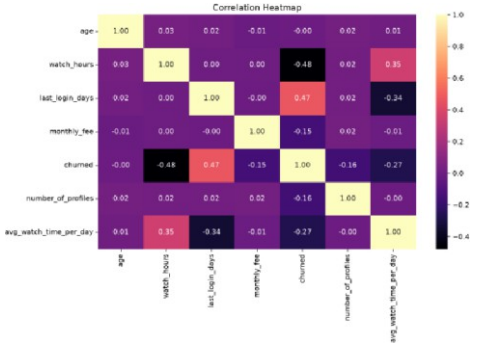


C. Correlation Heatmap

The correlation heatmap is used to examine the relationships between numerical attributes in the dataset. Different colours represent the strength and direction of correlations between features

```python
# GRAPH 2: CORRELATION HEATMAP
numeric_df = df.select_dtypes(include=np.number).drop(columns=["customer_id"], errors="ignore")
plt.figure(figsize=(10, 7))
sns.heatmap(numeric_df.corr(), annot=True, cmap="magma", fmt=".2f")
plt.title("Correlation Heatmap")
plt.tight_layout()
plt.show()
```

Output:



V. MACHINE-LEARNING METHODS

Five machine learning classification algorithms were implemented to predict customer churn using customer demographic information, subscription details, and viewing behaviour. The selected models are Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, and Support Vector Machine (SVM). Their performance was evaluated using accuracy, precision, recall, F1-score, and confusion matrices to identify the best-performing model for customer churn prediction.

A. Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for binary classification. In this project, it is used to predict whether a customer is likely to churn or remain subscribed. The model learns patterns from customer attributes and classifies customers into churn and non-churn categories. Its performance is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix.

```
#LOGISTIC REGRESSION

lr = LogisticRegression(max_iter=1000)
lr.fit(X_train, y_train)

y_pred_lr = lr.predict(X_test)

accuracy_lr = accuracy_score(y_test, y_pred_lr)
precision_lr = precision_score(y_test, y_pred_lr, zero_division=0)
recall_lr = recall_score(y_test, y_pred_lr, zero_division=0)
f1_lr = f1_score(y_test, y_pred_lr, zero_division=0)

print("\nLogistic Regression Results")
print("Accuracy:", round(accuracy_lr * 100, 2), "%")
print("Precision:", round(precision_lr, 3))
print("Recall:", round(recall_lr, 3))
print("F1-Score:", round(f1_lr, 3))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred_lr))
```

Fig 2: Implementation of the Logistic Regression Model for Customer Churn Prediction

B. K-Nearest Neighbours (KNN)

K-Nearest Neighbors (KNN) is a supervised machine learning algorithm used for classification. In this project, it predicts whether a customer will churn by comparing their data with similar customers in the dataset. The model uses the majority class of the nearest neighbors to classify customers into churn and non-churn categories. Its performance is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix.

```
# K-NEAREST NEIGHBORS (KNN)

knn = KNeighborsClassifier(n_neighbors=10)
knn.fit(X_train, y_train)

y_pred_knn = knn.predict(X_test)

accuracy_knn = accuracy_score(y_test, y_pred_knn)
precision_knn = precision_score(y_test, y_pred_knn, zero_division=0)
recall_knn = recall_score(y_test, y_pred_knn, zero_division=0)
f1_knn = f1_score(y_test, y_pred_knn, zero_division=0)

print("\nK-Nearest Neighbors Results")
print("Accuracy:", round(accuracy_knn * 100, 2), "%")
print("Precision:", round(precision_knn, 3))
print("Recall:", round(recall_knn, 3))
print("F1-Score:", round(f1_knn, 3))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred_knn))
```

Figure: Implementation of the K-Nearest Neighbours (KNN) Model for Customer Churn Prediction

C. Decision Tree

A Decision Tree is a supervised machine learning algorithm that uses a tree-like structure to make predictions. In this project, it predicts customer churn by analysing customer details, subscription information, and viewing behaviour. The model uses decision rules to classify customers into churn and non-churn categories. Its performance is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix.

```
# DECISION TREE

dt = DecisionTreeClassifier(criterion="entropy", max_depth=5, random_state=42)
dt.fit(X_train, y_train)

y_pred_dt = dt.predict(X_test)

accuracy_dt = accuracy_score(y_test, y_pred_dt)
precision_dt = precision_score(y_test, y_pred_dt, zero_division=0)
recall_dt = recall_score(y_test, y_pred_dt, zero_division=0)
f1_dt = f1_score(y_test, y_pred_dt, zero_division=0)

print("\nDecision Tree Results")
print("Accuracy:", round(accuracy_dt * 100, 2), "%")
print("Precision:", round(precision_dt, 3))
print("Recall:", round(recall_dt, 3))
print("F1-Score:", round(f1_dt, 3))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred_dt))
```

Fig 3: Implementation of the Decision Tree Model for Customer Churn Prediction

D. Random Forest

Random Forest is a supervised machine learning algorithm that combines multiple decision trees to make predictions. In this project, it predicts customer churn by analysing customer demographics, subscription details, and viewing behaviour. The algorithm combines the predictions of multiple trees to improve classification performance and reduce overfitting. Its performance is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix.

```
#----------------------------------------
#Decision Tree Classifier
#----------------------------------------

print("\n--- DECISION TREE CLIASSIFIER ---")
# Feature for Decision Tree
X = classification_data[["Model Year"]]
y = classification_data["EV Type"]
X_train, X_test, y_train, y_test = train_test_split(X, y,test_size=0.2,random_state=95,stratify=y)
tree_model = DecisionTreeClassifier(max_depth=5,random_state=95)
tree_model.fit(X_train, y_train)
y_pred = tree_model.predict(X_test)
acc = accuracy_score(y_test, y_pred)
prec = precision_score(y_test, y_pred, zero_division=0)
rec = recall_score(y_test, y_pred, zero_division=0)
f1 = f1_score(y_test, y_pred, zero_division=0)

print("\nDecision Tree Classifier Results")
print(f"Accuracy: {acc:.4f}")
print(f"Precision: {prec:.4f}")
print(f"Recall: {rec:.4f}")
print(f"F1 Score: {f1:.4f}")
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues")
plt.xlabel("Predicted Vehicle Type")
plt.ylabel("Actual Vehicle Type")
plt.title("Decision Tree Confusion Matrix")
plt.show()
```

Fig 4: Implementation of the Random Forest Model for Customer Churn Prediction

E. Support Vector Machine (SVM)

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification. In this project, it predicts whether a customer will churn or remain subscribed based on customer details, subscription information, and viewing behaviour. The model identifies a decision boundary that separates customers into churn and non-churn categories. Its performance is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix.

```
# SUPPORT VECTOR MACHINE (SVM)

svm = SVC(kernel="linear")
svm.fit(X_train, y_train)

y_pred_svm = svm.predict(X_test)

accuracy_svm = accuracy_score(y_test, y_pred_svm)
precision_svm = precision_score(y_test, y_pred_svm, zero_division=0)
recall_svm = recall_score(y_test, y_pred_svm, zero_division=0)
f1_svm = f1_score(y_test, y_pred_svm, zero_division=0)

print("\nSupport Vector Machine Results")
print("Accuracy:", round(accuracy_svm * 100, 2), "%")
print("Precision:", round(precision_svm, 3))
print("Recall:", round(recall_svm, 3))
print("F1-Score:", round(f1_svm, 3))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred_svm))
```

Fig 5: Implementation of the Support Vector Machine (SVM) Model for Customer Churn Prediction.

F. Model Evaluation

The five machine learning classification models were evaluated using accuracy, precision, recall, and F1-score. Confusion matrices were also used to examine the models' prediction results. These evaluation metrics help compare the performance of Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, and Support Vector Machine (SVM). The model with the highest accuracy was identified as the best-performing model based on the experimental results**.**

VI. EXPERIMENTAL DESIGN AND EVALUATION

The experimental design was established to evaluate five machine learning classification algorithms on the Netflix Customer Churn dataset. The dataset was preprocessed to handle missing values, remove duplicate records, and convert categorical attributes into numerical features. The data was divided into training and testing sets using an 80:20 ratio. Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, and Support Vector Machine (SVM) were trained to predict customer churn. Their performance was compared using accuracy, precision, recall, F1-score, and confusion matrices to identify the best-performing model.

A. Experimental Setup

The Netflix Customer Churn dataset was used to conduct the experiments. Data preprocessing included removing duplicate records, handling missing values, and converting categorical features into numerical form**.** The customer ID and churn status were separated from the input features, with churn status used as the target variable. The dataset was divided into 80% training data and 20% testing data. Feature scaling was performed using MinMaxScaler, and five classification models were trained and evaluated to predict customer churn**.**

B. Classification Evaluation

The five machine learning models were evaluated using accuracy, precision, recall, and F1-score. Accuracy measures the proportion of correctly predicted customer churn outcomes. Precision indicates how many customers predicted to churn actually churned, while recall measures how many actual churn cases were correctly identified. The F1-score combines precision and recall into a single measure. A confusion matrix was also used to analyse correct and incorrect predictions. These metrics provide a basis for comparing the performance of all five classification models.

C. Comparative Evaluation

The five machine learning models were compared based on their accuracy, precision, recall, and F1-score. These metrics were used to assess each model's ability to predict customer churn correctly. The results help identify the strengths and weaknesses of each algorithm. The model with the highest accuracy was selected as the best-performing model based on the experimental results. This comparison provides insight into the effectiveness of different machine learning algorithms for Netflix customer churn prediction**.**

VII. RESULTS AND DISCUSSION

A. Logistic Regression

Problem Addressed: Can Logistic Regression predict whether a Netflix customer will churn or remain subscribed?

Logistic Regression was used to predict customer churn based on customer demographics, subscription details, and viewing behaviour. The model classifies customers into churn and non-churn categories. Its performance was evaluated using accuracy, precision, recall, and F1-score. The evaluation results help determine how effectively Logistic Regression predicts customer churn**.**

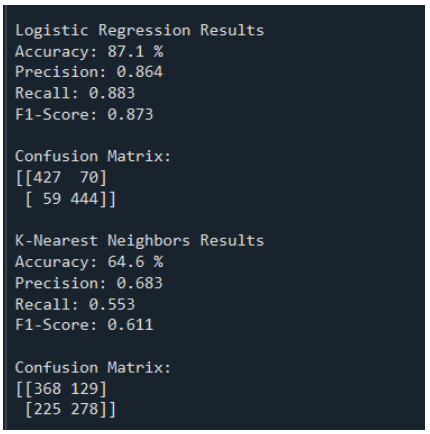


Fig 6: Performance Evaluation Output of the Logistic Regression Model for Netflix Customer Churn Prediction

B. K-Nearest Neighbors (KNN)

Problem Addressed: Can the K-Nearest Neighbors (KNN) algorithm predict whether a Netflix customer will churn or remain subscribed?

The K-Nearest Neighbors (KNN) algorithm was used to predict customer churn by comparing customer attributes with those of similar customers in the dataset. The model classifies customers based on the majority class among their nearest neighbors. Its performance was evaluated using accuracy, precision, recall, and F1-score to determine its effectiveness in predicting customer churn**.**

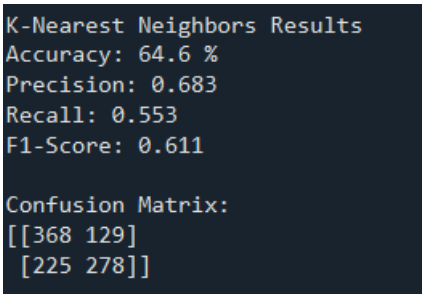


Fig 7: Performance Evaluation Output of the K-Nearest Neighbours (KNN) Model for Netflix Customer Churn Prediction

C. Decision Tree

Problem Addressed: Can the Decision Tree algorithm predict whether a Netflix customer will churn or remain subscribed?

The Decision Tree algorithm was used to predict customer churn by analysing customer demographics, subscription details, and viewing behaviour. It applies a set of decision rules to classify customers into churn and non-churn categories. The model's performance was evaluated using accuracy, precision, recall, and F1-score to measure its effectiveness in predicting customer churn.

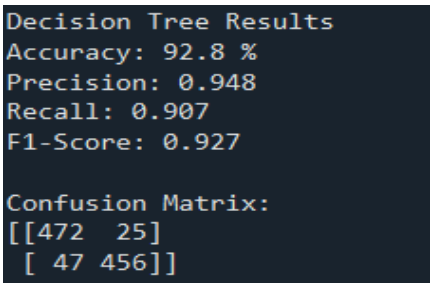


Fig 8: Performance Evaluation Output of the Decision Tree Model for Netflix Customer Churn Prediction

D. Random Forest

Problem Addressed: Can the Random Forest algorithm predict whether a Netflix customer will churn or remain subscribed?

The Random Forest algorithm was used to predict customer churn by combining the predictions of multiple decision trees. It analyses customer demographics, subscription details, and viewing behaviour to classify customers into churn and non-churn categories. The model's performance was evaluated using accuracy, precision, recall, and F1-score to assess its effectiveness in predicting customer churn**.**

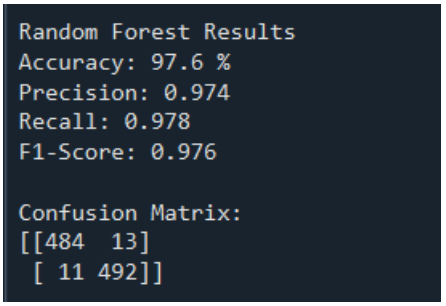


Fig 9: Performance Evaluation Output of the Random Forest Model for Netflix Customer Churn Prediction.

E. Support Vector Machine (SVM)

Problem Addressed: Can the Support Vector Machine (SVM) algorithm predict whether a Netflix customer will churn or remain subscribed?

The Support Vector Machine (SVM) algorithm was used to predict customer churn by identifying a decision boundary that separates churn and non-churn customers. It analyses customer information, subscription details, and viewing behaviour to classify customers. The model's performance was evaluated using accuracy, precision, recall, and F1-score to assess its effectiveness in predicting customer churn.

```
Support Vector Machine Results
Accuracy: 88.3 %
Precision: 0.878
Recall: 0.891
F1-Score: 0.885

Confusion Matrix:
[[435  62]
 [ 55 448]]
```

Fig 10: Performance Evaluation Output of the Support Vector Machine (SVM) Model for Netflix Customer Churn Prediction

F. Comparison of Model Performance

The performance of the five machine learning models was compared using accuracy, precision, recall, and F1-score. This comparison helps identify the strengths and weaknesses of Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, and Support Vector Machine (SVM). The model evaluation results can be used to determine which algorithm performs best for Netflix customer churn prediction. The final model is selected based on the actual evaluation results.

```
Comparison of All Models:
                Model  Accuracy  Precision  Recall  F1-Score
0  Logistic Regression     0.871      0.864   0.883     0.873
1                  KNN     0.646      0.683   0.553     0.611
2        Decision Tree     0.928      0.948   0.907     0.927
3        Random Forest     0.976      0.974   0.978     0.976
4                  SVM     0.883      0.878   0.891     0.885

Model with Highest Accuracy:
Model       Random Forest
Accuracy            0.976
Name: 3, dtype: object
```

Fig 11: Comparative Performance Analysis of Five Machine Learning Models for Netflix Customer Churn Prediction

Graph:

The bar chart compares the accuracy of five machine learning models. Random Forest achieved the highest accuracy at approximately 0.98, followed by Decision Tree at 0.93, SVM at 0.88, and Logistic Regression at 0.87. K-Nearest Neighbors (KNN) recorded the lowest accuracy at approximately 0.65. Based on this comparison, Random Forest performed best among the five models.

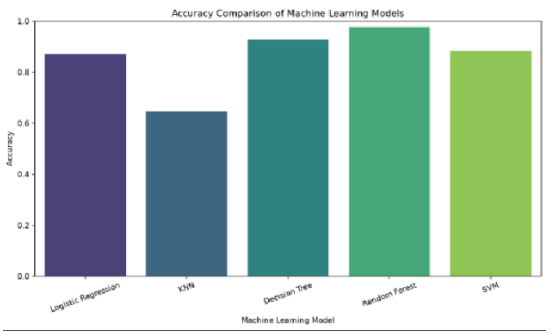
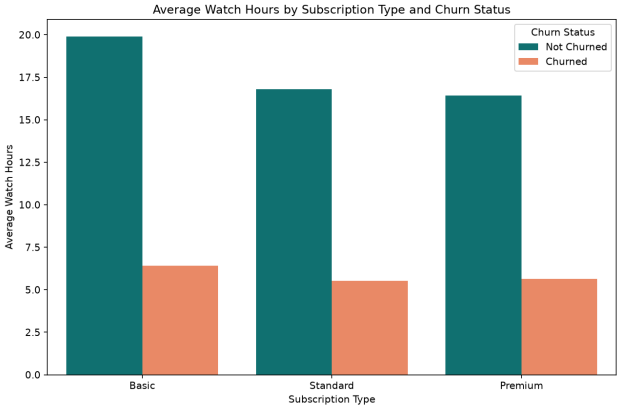


Fig 12: Accuracy Comparison of Five Machine Learning Models for Netflix Customer Churn Prediction

```python
# GRAPH 5 : AVERAGE WATCH TIME BY SUBSCRIPTION TYPE AND CHURN STATUS
plt.figure(figsize=(9, 6))
sns.barplot(
    data=df,
    x="subscription_type",
    y="watch_hours",
    hue="churned",
    palette=["teal", "coral"],
    errorbar=None
)
plt.title("Average Watch Hours by Subscription Type and Churn Status")
plt.xlabel("Subscription Type")
plt.ylabel("Average Watch Hours")
plt.legend(
    title="Churn Status",
    labels=["Not Churned", "Churned"]
)
plt.tight_layout()
plt.show()
```



VIII. Overall Discussion

The five machine learning algorithms were evaluated to predict customer churn in the Netflix dataset. The models were compared using accuracy, precision, recall, and F1-score. Among the five classifiers, Random Forest achieved the highest accuracy, followed by Decision Tree, SVM, Logistic Regression, and K-Nearest Neighbors (KNN).

Overall, the results indicate that machine learning can help identify customers who may discontinue their subscriptions. Random Forest demonstrated the best accuracy among the evaluated models. Further improvements may be achieved by including additional relevant customer features, tuning model parameters, and ensuring proper data preprocessing to improve prediction reliability.

IX. CONCLUSION

This project focuses on predicting customer churn using machine learning techniques on the Netflix Customer Churn dataset. The main objective is to identify customers who may discontinue their subscriptions by analysing customer demographics, subscription details, and viewing behaviour. Data preprocessing was performed to handle missing values, remove duplicate records, and prepare the dataset for model training.

Data visualization techniques were used to understand customer characteristics and explore patterns related to churn. The churn distribution graph, customer age histogram, and correlation heatmap helped analyse the dataset and understand relationships between different attributes. These visualizations provided useful insights before applying machine learning algorithms.

Five machine learning classification algorithms—Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, and Support Vector Machine (SVM)—were implemented and evaluated. Their performance was compared using accuracy, precision, recall, and F1-score. Based on the accuracy comparison graph, Random Forest achieved the highest accuracy, while KNN recorded the lowest accuracy among the five models.

Overall, this project demonstrates how machine learning can support customer churn prediction and help businesses identify customers who may stop using their services. The results can assist in developing customer retention strategies and improving customer satisfaction. In the future, the project can be improved through feature engineering, hyperparameter tuning, and testing additional machine learning techniques to achieve more reliable predictions.

X. REFERENCES

1. Dataset Link: DATA_SET_LINK
2. LinkedIn: LINKEDIN_LINK
3. GitHub: GITHUB_LINK